September 25, 2026

Agenda (1/2)

- Context: the ZigBee standard and its relation to IEEE 802.15.4
- General characteristics and frequency bands
- Device roles in a ZigBee network
- Network topologies: star, tree and mesh
- Addressing and network formation
- Protocol stack and application profiles

Agenda (2/2)

- Energy consumption and use cases
- Summary

ZigBee and the IEEE 802.15.4 Standard

- ZigBee is a network specification maintained by the ZigBee Alliance (now part of the Connectivity Standards Alliance), built on the physical and medium access control layers defined by IEEE 802.15.4
- IEEE 802.15.4 defines a low-rate wireless personal area network (LR-WPAN), concerned above all with low consumption and low cost, and not with high data rates
- ZigBee adds to that base the network (NWK) and application support (APS) layers, as well as a set of standardised application profiles
- The result is a complete stack, designed for sensor and actuator networks with many nodes, battery-powered over long periods

General Characteristics of ZigBee (1/2)

- Very low energy consumption, with extremely low duty cycles
- Low data rate, between 20 and 250 kbps, depending on the frequency band used
- Short range, typically between 75 and 100 metres per hop
- Fast network association time, in the order of 30 milliseconds

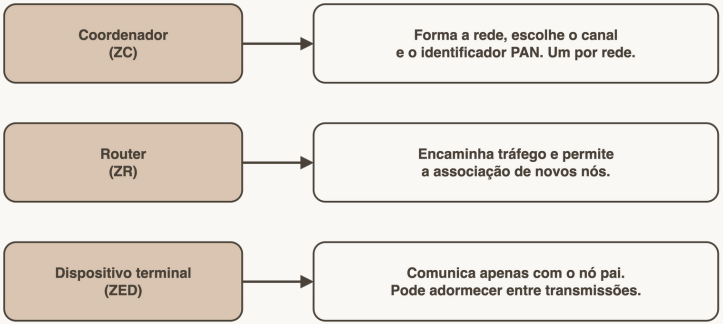
General Characteristics of ZigBee (2/2)

- Support for small and large networks: up to 65000 devices in theory, although in practice the figure is around 240 units
- Low implementation cost, as it is an open specification protocol

Frequency Bands and Channels

- Channel 0: the 868 MHz band, used in Europe, with a single channel available
- Channels 1 to 10: the 915 MHz band, used in the United States and Australia
- Channels 11 to 26: the 2.4 GHz band, an ISM band available worldwide, with 16 channels
- In total, the standard defines 27 channels across three frequency bands, which allows the channel choice to be adapted to the spectrum rules of each region and interference with other wireless networks to be avoided

Device Types and Their Functions



Só o dispositivo terminal pode adormecer; é o único adequado a alimentação por bateria

Device Roles: the Coordinator (1/2)

- The ZigBee Coordinator (ZC) is a full function device (FFD, in IEEE 802.15.4 terminology) and there is exactly one per network
- It is responsible for starting the network, choosing the channel and the PAN identifier
- It stores the information relating to the network and to all devices that associate with it

Device Roles: the Coordinator (2/2)

- All remaining devices communicate, directly or indirectly, with the coordinator
- It may perform routing functions and act as a bridge to other networks

Device Roles: Router and End Device (1/2)

- The ZigBee Router (ZBR) is also an FFD, but optional in the network
- It forwards messages between nodes, acting as the backbone of the network, and extends coverage beyond the direct range of the coordinator
- It also manages the allocation and release of local addresses to the devices that associate with it

Device Roles: Router and End Device (2/2)

- The ZigBee End Device (ZED) is a reduced function device (RFD), optimised for low energy consumption
- It is the cheapest device type in the network and typically where a sensor or the equipment to be controlled is installed

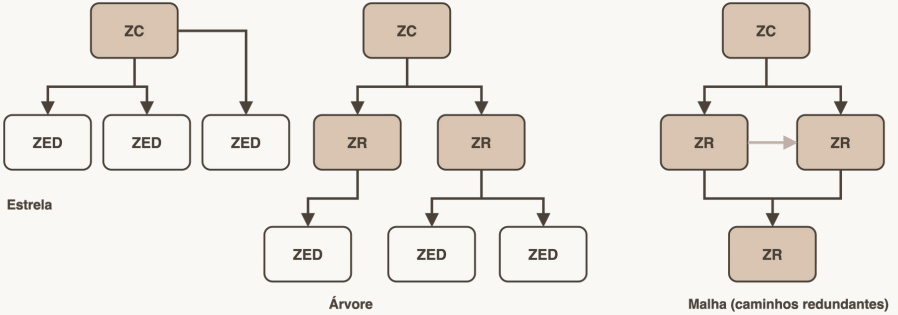
FFD, RFD and Modes of Operation (1/2)

- Full Function Device (FFD): it may take on any role in the network, including coordinator or router, and communicate with any other device
- Reduced Function Device (RFD): it has limited capabilities, communicates only with a single parent device and is intended for simple low-consumption applications
- Devices alternate between two states, active and sleeping, which saves energy when there is no data to transmit

FFD, RFD and Modes of Operation (2/2)

- The network may operate in beacon mode, with periodic synchronisation between nodes, or in non-beacon mode, without that synchronisation signal
- Traffic observed in a ZigBee network may be intermittent, repetitive or periodic, depending on the application

The Three ZigBee Topologies



Star Topology

- A single central coordinator concentrates all network communication
- All remaining devices communicate only with that coordinator, never directly with one another
- It is the simplest topology to implement and manage
- Its limitations are that the coordinator is a single point of failure and that network coverage is confined to the radio range of that single central node

Tree Topology (Cluster-Tree)

- The coordinator and the routers may emit periodic beacons, which synchronise child devices and organise medium access
- The network is organised in a hierarchical structure of parent-child relationships, starting from the coordinator
- Routers extend the reach of the network, connecting successive branches or clusters from the central node
- It is an intermediate topology between the simplicity of the star and the robustness of the mesh, but retains some dependence on the established hierarchy

Mesh Topology

- In a mesh network, regular beacons are not permitted
- Devices communicate with one another by direct, peer-to-peer transmissions, without depending on a single central node
- Each device may forward messages on behalf of others, creating multiple possible paths between source and destination
- This path redundancy makes the network more robust to individual node failures and is the reason ZigBee is particularly suited to sensor networks and to larger-scale home automation applications

Comparison of Topologies

- Star: simplest and with lowest latency to reach the coordinator, but with no redundancy and limited coverage
- Tree: greater reach than the star, good routing predictability, but still dependent on the parent-child hierarchy
- Mesh: greater robustness and greater aggregate reach, at the cost of higher routing and network management complexity
- The choice of topology depends on the intended trade-off between simplicity, coverage, reliability and management complexity

Device Addressing (1/2)

- Two or more devices communicating on the same physical channel constitute a WPAN, which always includes at least one PAN coordinator
- Each independent PAN chooses a unique identifier, the PAN ID
- Every device has a 64-bit extended address, inherited from IEEE 802.15.4, which may be used for direct communication within the PAN

Device Addressing (2/2)

- On association, the coordinator assigns each device a 16-bit short address, more efficient for routine use in the network
- Up to 256 sub-addresses may further be allocated to subunits of the same device

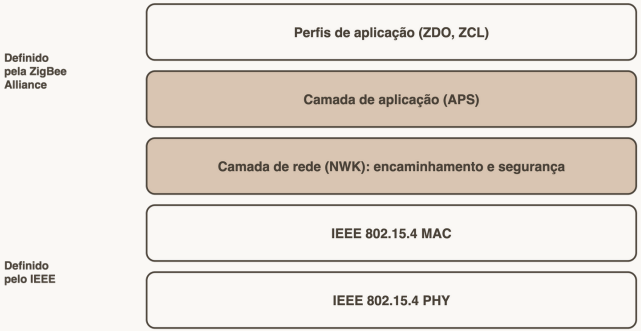
Network Formation and Associated Mechanisms (1/2)

- Stochastic addressing: each device is assigned a random address, which is then announced, with a conflict resolution mechanism in place; parent nodes need not maintain a table of already assigned addresses
- Link management: each node records the quality of the link to each neighbour, using it as a routing cost
- Frequency agility: a node suffering interference reports it to a channel manager, which may select another channel for the whole network

Network Formation and Associated Mechanisms (2/2)

- Asymmetric links: each node has its own transmit power and sensitivity, so paths between two nodes may have different quality in each direction
- Power management: coordinators and routers are typically mains-powered, while end devices run on batteries

Protocol Stack and Responsibilities



ZigBee Protocol Stack (1/2)

- Physical layer (PHY) and medium access control layer (MAC), inherited directly from IEEE 802.15.4
- Network layer (NWK): responsible for topology, routing and address assignment
- Application support layer (APS): it bridges the network layer and the applications, managing bindings between devices

ZigBee Protocol Stack (2/2)

- ZigBee Device Object (ZDO): it defines the role of the device in the network (coordinator, router or end device), initiates and responds to binding requests, and establishes security relationships between devices
- Application layer: it hosts the application profiles and the objects representing the concrete devices

Application Profiles and ZigBee Device ID (1/2)

- Application profiles define a common language for data exchange, specifying the services offered and ensuring interoperability between devices from different manufacturers
- Public profiles are made available by the ZigBee Alliance, contain standardised device descriptions and a unique identifier licensed by the Alliance
- Manufacturer-specific profiles are intended for proprietary applications, developed privately; they must nonetheless use a profile identifier assigned by the Alliance, and commercial products implementing them must undergo network capability testing

Application Profiles and ZigBee Device ID (2/2)

- Each endpoint of a device includes a profile identifier and a device identifier (device ID), representing something physical and concrete: a lamp, a temperature sensor, a thermostat, for instance
- Device IDs lie in the range between 0x0000 and 0xFFFF and their meaning depends on the profile to which they belong; the Home Automation Profile is an example of a profile defining its own set of device IDs

Energy Consumption (1/2)

- ZigBee was designed from the outset for very low duty cycles, with devices spending most of their time in sleep mode
- End devices wake only periodically to check with the parent device whether there are pending messages, which allows them to run for years on a single battery

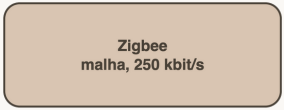
Energy Consumption (2/2)

- Coordinators and routers must remain permanently active in order to forward traffic, and are therefore normally mains-powered
- The fast network association time, in the order of 30 milliseconds, also contributes to reducing consumption, since devices need not remain active for long in order to join the network

Typical Use Cases

- Home automation: lighting, thermostats, locks and other devices that need to communicate with one another and with a central control hub
- Wireless sensor networks: environmental, industrial and agricultural monitoring, with many nodes spread over a wide area
- Smart buildings and commercial lighting, where the mesh topology allows large areas to be covered without a single central access point
- In all these scenarios, the decisive factor is not data rate, but the energy autonomy of the devices and the scalability of the network to hundreds of nodes

ZigBee Compared with Other Short-Range Technologies



Consumo muito baixo
Alcance alargado por salto



Consumo baixo
Sem encaminhamento entre nós



Consumo elevado
Infraestrutura existente

Advantages and Limitations (1/2)

- Advantages: low energy consumption; security and application support services built on IEEE 802.15.4; low cost of modules and chips; mesh topology, with no need for a central hub; high reliability even in adverse conditions
- Limitations: short range, which may be insufficient in larger buildings; low data rate, poorly suited to bulk transfers; smaller market adoption than other IoT protocols, with an impact on interoperability between manufacturers; security mechanisms less robust than those of more recent alternatives

Advantages and Limitations (2/2)

- Compared with technologies such as Bluetooth Low Energy, ZigBee is distinguished by its native mesh topology and by an established presence in home automation applications, although BLE has been gaining ground in energy consumption and in presence on mobile devices

Summary (1/2)

- ZigBee is a low-power mesh network stack, built on IEEE 802.15.4, aimed at sensor and actuator networks with many nodes
- It does not seek high data rates or long range; it seeks energy efficiency, low cost and scalability to hundreds of devices
- The coordinator, router and end device roles, combined with the star, tree and mesh topologies, give flexibility to adapt the network to different deployment scenarios

Summary (2/2)

- Application profiles ensure that devices from different manufacturers can interoperate, which is essential in home automation applications
- It coexists with other short-range technologies, such as Bluetooth Low Energy, the choice resting on the specific nature of each application

Sources

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Thank you.

Questions and comments